TETHER: Supplementary Material

Anonymous Authors

***Abstract*— This document reports measurements that did not fit the page limit. It adds no claim to the main paper. Two of its sections (Sec. III and Sec. VII) contain results that *qualify* the main claims rather than strengthen them, and they are included for that reason.**

I. Complete Matched-Baseline Closure

Table I lists all fifteen policies of the D2 closure. The main paper reports a representative subset. Every row shares the bounded state machine, the raw re-anchor schedule, the SEA-RAFT alignment, the support definition and the fusion form; only the blending weight differs, and all 225 reports (15 methods $\times$ 5 backbones $\times$ 3 sequences) come from a single run under one frozen protocol.

TABLE I

Full SCARED-C D2 matched temporal-policy closure. Raw EPE is 0.276202. Positive reduction denotes improvement.

Method	EPE	Reduction
TETHER $H = 6$	**0.264656**	4.18%
TETHER $H = 8$	0.264912	4.09%
TETHER $H = 4$	0.265028	4.05%
TETHER $H = 2$	0.267265	3.24%
TETHER $H = 1$	0.269001	2.61%
Fixed $w = 0.5$	0.274562	0.59%
EMA2	0.274562	0.59%
Fixed $w = 0.3$	0.274852	0.49%
Fixed $w = 0.2$	0.275229	0.35%
EMA3	0.275386	0.30%
Fixed $w = 0.1$	0.275684	0.19%
Warped raw previous	0.276298	−0.03%
FB confidence weight	0.282182	−2.17%
Warped recurrent $w = 1$	0.286035	−3.56%
Continuous recurrence	0.449349	−62.69%
Oracle raw/memory	0.242223	12.30%

a) An identity worth stating.: `EMA2` and `Fixed` $w = 0.5$ are the same policy under two names, and `EMA3` is $w = 2/3$: under a recurrent state a fixed convex weight *is* an exponential moving average. The baseline family is therefore five fixed weights $w \in \{0.1, 0.2, 0.3, 0.5, 2/3\}$ over the shared bounded state, not two independent families. We state this because two nominally different baselines producing an identical number otherwise reads as a bug.

II. The Oracle Ceiling Is Not the One We Report Against

The oracle of the main paper selects, per pixel, the better of raw disparity and aligned memory. It is therefore the ceiling of $w \in \{0, 1\}$, whereas TETHER acts on $w \in [0, 1]$, whose ceiling is the continuous oracle $w^\star = \mathrm{clip}\big((g - d^{\mathrm{raw}})/(\widetilde{m} - d^{\mathrm{raw}}), 0, 1\big)$.

TABLE II

Binary versus continuous oracle on D2, per backbone.

Backbone	raw	TETHER	$\{0,1\}$	$[0,1]$
S^2M^2-S	.2949	.2850	.2634	.2610
RAFT-Stereo	.3022	.2914	.2687	.2658
Stereo Anywhere	.3189	.2964	.2747	.2720
CREStereo	.2943	.2832	.2610	.2582
Fast-FoundationStereo	.3037	.2930	.2715	.2688

The correction is real but small. Only 6.1% of D2 pixels have the ground truth bracketed by the two candidates (11–14% on D7), so the two ceilings nearly coincide and the recovered fraction falls from 32.9% to 30.1%. We report it because the published figure is measured against a ceiling the method cannot actually be limited by, not because it changes the conclusion.

III. Motion-Compensated Multi-Horizon Fidelity

$\mathrm{TEPE}_{\mathrm{corr}}$ and OPW require a correspondence field, supplied here from the same SEA-RAFT flows used for warping. Table III gives the relative change after refinement.

On held-out D7 both improve in all $20/20$ backbone–horizon cases. On development D2 they disagree: OPW improves everywhere while $\mathrm{TEPE}_{\mathrm{corr}}$ degrades everywhere.

This rules out a convenient explanation. One might attribute a rising temporal-error metric to a grid metric penalising legitimate scene motion, but $\mathrm{TEPE}_{\mathrm{corr}}$ *is* motion compensated and still degrades on D2. The two ask different questions: OPW asks whether consecutive predictions agree with each other, $\mathrm{TEPE}_{\mathrm{corr}}$ whether the predicted temporal change matches the true one. On the split used to select the operating point, the module therefore becomes more self-consistent while tracking the true change less faithfully. We report this against our own interest.

IV. Forensics of the Support Violation

The main paper states that an unmasked output lets zero-padded out-of-bounds memory become an invalid prediction. The supporting measurement is below: every offending pixel has an aligned memory of *exactly* zero, and the offenders arrive in bursts no longer than the state lifetime, so bounded recurrence contains the damage it cannot prevent.

Because an invalid prediction on valid support is penalised rather than discarded, at 10^4 mm per pixel, a few hundred pixels out of 38.8 million decide the external-domain metric-depth verdict.

TABLE III

$\mathrm{TEPE}_{\mathrm{corr}}$ (TOP) AND OPW (BOTTOM), RELATIVE CHANGE AFTER REFINEMENT. NEGATIVE IS BETTER.

Split	Backbone	k=1	k=2	k=4	k=8
$\mathrm{TEPE}_{\mathrm{corr}}$					
D2	S^2M^2-S	+24.5%	+40.1%	+12.9%	+19.7%
D2	RAFT-Stereo	+24.8%	+41.1%	+16.6%	+23.8%
D2	Stereo Anywhere	+14.5%	+31.1%	+8.5%	+16.4%
D2	CREStereo	+22.2%	+39.7%	+15.3%	+24.3%
D2	Fast-FoundationStereo	+35.9%	+52.3%	+21.1%	+27.8%
D7	S^2M^2-S	-23.7%	-21.0%	-17.4%	-12.7%
D7	RAFT-Stereo	-12.2%	-6.1%	-5.6%	-5.7%
D7	Stereo Anywhere	-10.3%	-4.8%	-5.3%	-5.7%
D7	CREStereo	-13.6%	-6.8%	-6.7%	-6.4%
D7	Fast-FoundationStereo	-10.9%	-4.9%	-5.6%	-6.2%
OPW					
D2	S^2M^2-S	-8.7%	-3.5%	-1.8%	-1.2%
D2	RAFT-Stereo	-8.5%	-3.4%	-1.7%	-1.1%
D2	Stereo Anywhere	-12.9%	-5.8%	-3.0%	-1.8%
D2	CREStereo	-9.5%	-3.8%	-1.8%	-1.2%
D2	Fast-FoundationStereo	-7.3%	-3.1%	-1.6%	-1.2%
D7	S^2M^2-S	-29.8%	-22.8%	-13.5%	-7.6%
D7	RAFT-Stereo	-18.5%	-10.2%	-5.1%	-3.4%
D7	Stereo Anywhere	-18.1%	-9.9%	-5.1%	-3.4%
D7	CREStereo	-20.3%	-11.1%	-5.9%	-3.8%
D7	Fast-FoundationStereo	-18.5%	-10.0%	-5.2%	-3.6%

TABLE IV

INVALID REFINED PIXELS UNDER THE UNMASKED OUTPUT, AND THE STATE AGE AT WHICH THEY OCCUR.

	Vid10_Liver_Med	Vid11_Liver_High
Support pixels	38,828,160	38,439,360
Invalid (unmasked)	300	265
Invalid (support-confined)	**0**	**0**
Frames affected	10	3
Aligned memory, min/max	0.0 / 0.0	0.0 / 0.0
Non-finite memory	0	0
State age 1/2/3/4	10/103/135/52	0/182/72/1

V. Per-Recording DRENDS Transfer

All 60 recording–backbone–metric cells improve. The per-recording pattern is set by the recording and not by the estimator: on Vid10 the EPE change is -5.7%, -5.2% and -5.0% for the seen and the two unseen backbones. That alignment, rather than the pooled mean, is what supports backbone independence.

VI. Stage-Wise Runtime

NVIDIA H100 PCIe, PyTorch 2.5.1+cu121, 144×180 evidence grid, 200 timed iterations after 30 warmup iterations.

Two observations. The 177k-parameter head costs 1.03 ms, 3% of the overhead: "compact head" is true but not the operative cost, since 79% is the two frozen flow passes. And the reverse pass costs 13.19 ms while feeding only the forward–backward confidence cue, whose direct use as a fusion weight worsens EPE by 2.17% (Table I); removing it would cut overhead by 39% and is the most promising efficiency ablation we have not run.

TABLE V

ZERO-SHOT DRENDS, RELATIVE CHANGE AFTER REFINEMENT, ALL METRICS AND ALL FIVE RECORDINGS. NEGATIVE IS BETTER.

Backbone	Recording	EPE	Bad3	d.MAE	d.RMSE
RAFT-Stereo	Vid10	-5.7%	-11.3%	-3.8%	-3.4%
(seen)	Vid11	-3.4%	-21.9%	-2.5%	-1.5%
	Vid12	-8.4%	-22.7%	-6.7%	-6.0%
	Vid13	-4.3%	-14.2%	-3.2%	-3.0%
CREStereo	Vid10	-5.2%	-8.7%	-3.4%	-3.2%
(unseen)	Vid11	-3.7%	-22.8%	-2.7%	-1.8%
	Vid12	-6.3%	-18.5%	-4.8%	-4.3%
	Vid13	-5.2%	-10.4%	-3.6%	-3.3%
	Vid14	-5.6%	-26.5%	-4.4%	-4.1%
Fast-Found.	Vid10	-5.0%	-7.3%	-3.0%	-2.9%
(unseen)	Vid11	-3.2%	-20.4%	-2.3%	-1.4%
	Vid12	-5.3%	-10.3%	-4.4%	-3.6%
	Vid13	-4.3%	-10.9%	-3.0%	-2.9%
	Vid14	-4.7%	-27.4%	-3.8%	-3.5%

TABLE VI

PER-STAGE TEMPORAL OVERHEAD.

Stage	median (ms)	p95 (ms)
SEA-RAFT forward	13.222	13.289
SEA-RAFT reverse	13.191	13.268
Alignment	1.038	1.057
142-channel evidence	5.122	5.156
Fusion head	**1.027**	1.043
Total overhead	**33.601**	
Peak VRAM		**124.2 MB**

VII. Three-Dimensional Surface Consistency

SCARED-C provides no per-frame camera pose, so global reconstruction drift cannot be measured. We instead back-project each frame's disparity to a metric cloud and compare it against the previous frame's cloud pulled through the same SEA-RAFT correspondence: frame-to-frame surface stability, not reconstruction drift.

Excess over the GT floor falls on four backbones of five, for instance $0.0326 \rightarrow 0.0276$ mm on RAFT-Stereo, and rises slightly on Fast-FoundationStereo. The effect is real but of the order of 0.005 mm, which is why it does not appear in the paper.

The normal metric carries a caveat that removes its floor entirely: GT normals are *noisier* than every prediction ($5.86°$ against 3.74–$5.60°$), because the pseudo-ground-truth is sparse and central differences amplify it. Predictions smoother than the reference cannot be scored against that reference, so the normal column should be read as a relative comparison between raw and refined only.

Further assumptions a reader may wish to challenge: the principal point is taken at the image centre with $f_y = f_x$, since the SCARED-C manifest carries only f_x and the baseline; flow correspondence stands in for true scene-point correspondence, gated by forward–backward consistency at library defaults; and both clouds live in their own camera

TABLE VII

FRAME-TO-FRAME 3D SURFACE STABILITY ON D2, POOLED MEANS. GT IS THE FLOOR SET BY REAL SCENE AND CAMERA MOTION.

Backbone	raw	refined	GT
point-to-point distance [mm]			
S^2M^2-S	0.5016	0.4978	0.4758
RAFT-Stereo	0.5085	0.5034	0.4758
Stereo Anywhere	0.5060	0.4996	0.4758
CREStereo	0.5065	0.5004	0.4758
Fast-FoundationStereo	0.4890	0.4902	0.4758
surface-normal angle [deg]			
S^2M^2-S	4.4878	4.3241	5.8586
RAFT-Stereo	5.1443	4.7297	5.8586
Stereo Anywhere	5.4499	4.8516	5.8586
CREStereo	5.6006	4.9958	5.8586
Fast-FoundationStereo	3.7381	3.8742	5.8586

frames, so the measure mixes surface change with rigid camera motion, a component the GT floor absorbs but does not isolate.

VIII. LARGER TRAINING DOES NOT MOVE THE OPERATING POINT

An augmented-training run keeps the architecture and enlarges the data.

TABLE VIII

HELD-OUT D7, CANONICAL VERSUS AUGMENTED TRAINING.

Metric	Canonical	Augmented
Mean EPE	**.4693**	.4706
HUR	**.3965**	.4457
H^+	**.0148**	.0235
B^+	.0376	**.0450**
BUR	.4951	**.5060**
Degraded-frame fraction	**.2642**	.3341
Worst frame degradation	**.1063**	.3958

The augmented model performs more beneficial updates and substantially more harmful ones. Scale shifts intervention aggressiveness without improving held-out risk–gain balance.

IX. EXTERNAL STABILISER CONTEXT

BiDAStabilizer [1] is the closest published plug-in stabiliser that can actually be run. The main paper reports the pooled comparison; this section records how it was made fair and what it looks like sequence by sequence.

Our first attempt was not comparable at all, for two independent reasons: the published reproduction ran RAFT-Stereo *robust* while we ran RAFT-Stereo *middlebury*, and it was scored under a different support contract (raw EPE 0.3029 against 0.2762). Subtracting those numbers would have been meaningless. Both differences are removed by reading the stored NPZ boundary the reproduction already wrote — the exact RGB and the exact disparity BiDAStabilizer consumed — running our module on that same input, and scoring all three conditions against the same ground truth on one prediction-independent support: GT coverage $\cap$ raw validity, which neither method can influence. The one remaining difference is the one that matters and is reported rather than removed: BiDAStabilizer is bidirectional and consumes future frames.

TABLE IX

PER-SEQUENCE COMMON-SUPPORT COMPARISON, RELATIVE CHANGE AGAINST RAW (NEGATIVE IS BETTER). FRAMES AND SUPPORT ARE IDENTICAL WITHIN A ROW GROUP.

Sequence	Method	EPE	Bad1	Bad3
keyframe_2 (1033 fr.)	BiDAStabilizer	-1.45	-4.01	$\mathbf{-5.31}$
	TETHER	$\mathbf{-2.64}$	$\mathbf{-17.67}$	-4.18
keyframe_3 (1102 fr.)	BiDAStabilizer	$\mathbf{-2.32}$	$\mathbf{-2.99}$	$\mathbf{-27.99}$
	TETHER	$+0.27$	-2.34	-10.97
keyframe_4 (2114 fr.)	BiDAStabilizer	$+0.63$	-5.81	-6.82
	TETHER	$\mathbf{-6.53}$	$\mathbf{-17.17}$	$\mathbf{-16.52}$
Pooled (18.0M px)	BiDAStabilizer	-0.25	-4.83	-8.32
	TETHER	$\mathbf{-4.58}$	$\mathbf{-12.53}$	$\mathbf{-15.52}$

The pooled row favours TETHER on every metric, but two of the nine per-sequence cells do not, and one sequence favours BiDAStabilizer outright. On `keyframe_3` our EPE is 0.27% *worse* than raw while the bidirectional method improves it by 2.32%. We report this rather than pooling it away: with three sequences, a single sequence-level disagreement is well within what the split can produce, and a method that sees the future should be expected to win somewhere. The claim the comparison supports is the pooled one, and it is that a strictly causal module beats a bidirectional one on the same input, not that it does so everywhere.

X. ACQUISITION AUDIT OF THE COMPARISON SET

The main paper states that no published method in our category can be run head-to-head. That is a strong claim, so we record what was attempted, on which date, and with what response. Everything below was checked on 16 August 2026 from the same host and the same session; the same `curl` reached `huggingface.co` and `archive.org` throughout, so a failure below is the resource's, not our network's.

TABLE X

WHAT IS ACTUALLY OBTAINABLE. "CATEGORY" IS CAUSAL + BLACK-BOX + ATTACHED TO A FROZEN STEREO NETWORK, WHICH IS OUR SETTING.

Method	Venue	Code	Weights
StereoDiffusion [2]	MICCAI'24	none	none
Neural Disp. Refinement [3]	3DV'21	yes	deleted
Lai et al. [4]	ECCV'18	yes	host dead
TC-Stereo [5]	ECCV'24	yes	yes
BiDAStabilizer [1]	ECCV'24	yes	yes

a) StereoDiffusion.: The only published method in our exact category. Its repository has one commit, 7.43 KiB of git objects, and contains a `.gitignore`, a `LICENSE` and a twelve-byte `README.md`. `git ls-remote -heads` shows only `main`. There is no implementation to run.

b) Neural Disparity Refinement.: Same input contract as ours (RGB plus a black-box disparity), spatial only. The code clones and imports. Both released Google Drive identifiers return HTTP 404 from `drive.usercontent.google.com/download`, from `docs.google.com/uc`, *and* from the plain `drive.google.com/file/d/.../view` landing page. A file that exists but is access-restricted answers 403 or serves an interstitial; 404 on the landing page means the object is gone. Hugging Face model search returns nothing for "neural disparity refinement" or "disparity refinement"; the GitHub releases API for the official repository returns an empty list; Zenodo returns zero records; ModelScope has no such model.

c) Lai et al.: The natural task-agnostic baseline: model-agnostic and causal at inference. Its `pretrained_models/download_models.sh` points at `vllab.ucmerced.edu`, which answers 200 for the directory listing and 404 for every checkpoint. The Wayback CDX index holds the project page, its stylesheets and the 34 MB supplementary PDF, and zero captures of `ECCV18_blind_consistency.pth` or its optimiser file — web crawlers archive HTML and skip large binaries. No Hugging Face mirror and no GitHub release exists.

d) TC-Stereo.: Obtainable and loadable: we found a configuration that matches all three released checkpoints with zero missing and zero unexpected keys (16.7 M parameters). Its blocker is scientific. The temporal branch warps the previous disparity by a rigid 6-DoF reprojection assembled from per-frame camera pose, intrinsics and baseline. SCARED-C provides no per-frame pose, and deformable tissue violates the rigid-scene assumption independently of that. We therefore disable the temporal branch, which reduces the network to its single-frame stereo path, and report it as an integrated-architecture reference rather than as a temporal competitor. The run manifest records the disabled branch and this reason.

e) Deep Video Prior.: Excluded on definition rather than on availability: it performs per-video test-time training over an entire clip, so it is not causal under any configuration.

The practical consequence is that our comparison budget goes where a comparison is actually possible: BiDAStabilizer on a common support (Section IX), TC-Stereo as the integrated reference, and the matched-baseline closure of Section I, in which every degenerate temporal policy shares our alignment, our support contract and our evaluation code and differs only in the decision rule.

XI. Seed Study in Full

Pre-registered before any seed result existed. The seed is the only field that varies from the locked recipe; the training script is a sibling of `train.py` that reuses its guards and overrides only that field. Sequences are the resampling unit for the paired bootstrap, seeds are averaged inside each resample, and four things were prohibited in advance: selecting the best seed, re-tuning any threshold after seeing seed results, changing the checkpoint-selection rule, and using D7 for any decision other than the final reported numbers.

TABLE XI

Relative Bad1 reduction (%) per backbone over three seeds, with the paired sequence-level bootstrap interval and the fraction of resamples in which refinement failed to help.

Split	Backbone	s0	s1	s2	mean ± std	CI_{95}	p
D2	CREStereo	8.26	6.62	3.57	6.15 ± 2.38	$[0.69, 12.74]$	.000
D2	Fast-Found.	7.91	5.74	2.65	5.43 ± 2.64	$[0.63, 12.86]$	.000
D2	RAFT-Stereo	6.95	5.40	3.00	5.12 ± 1.99	$[-0.12, 12.43]$	.035
D2	S^2M^2-S	7.40	6.70	4.45	6.18 ± 1.54	$[-0.31, 12.32]$	.038
D2	StereoAnywhere	19.65	15.95	12.43	16.01 ± 3.61	$[13.89, 19.26]$	.000
D7	CREStereo	3.86	4.51	4.32	4.23 ± 0.33	$[2.68, 7.27]$	.000
D7	Fast-Found.	3.97	4.08	3.76	3.93 ± 0.16	$[1.60, 8.58]$	.000
D7	RAFT-Stereo	3.95	5.35	4.99	4.77 ± 0.73	$[3.27, 7.72]$	.000
D7	S^2M^2-S	5.66	6.25	6.05	5.99 ± 0.30	$[1.50, 13.17]$	.000
D7	StereoAnywhere	6.40	7.07	6.73	6.73 ± 0.33	$[5.66, 12.12]$	.000
Pooled, 35 sequences		5.79	6.00	5.28	5.69 ± 0.37	$[4.11, 8.02]$	.000

Two features of this table are worth stating plainly rather than leaving for a reader to find.

First, three D2 rows have intervals that touch or cross zero. That is a property of the split, not of the method: D2 contributes three sequences, and a three-sequence bootstrap is genuinely wide. We prefer reporting it to quietly resampling pixels, which would have produced narrow intervals with no statistical meaning. Held-out D7 contributes four sequences and every one of its rows is strictly positive at $p = .000$ on both metrics.

Second, the seed ordering differs between splits in a way that is not noise. Pooled per split, the D2 Bad1 reduction falls monotonically across seeds — 10.78, 8.67, 5.75 — while D7 gives 4.78, 5.47, 5.19, in which seed 0 is the *weakest*. Ten of ten per-backbone comparisons follow the same direction. Our checkpoint-selection rule picks best D2 fused EPE, so seed 0 is by construction the run that looked best on D2; the D2 margin therefore carries selection optimism and the held-out margin does not.

XII. A Metric Field We Do Not Report

FrameDegradation's CatastrophicFraction was numerically identical to PositiveFraction on every inspected row. The cause is definitional: the two are $\Pr(D_t > \tau_{\mathrm{cat}})$ and $\Pr(D_t > 0)$, and the catastrophic threshold defaults to $\tau_{\mathrm{cat}} = 0$, which the evaluators never override. At that setting they coincide by construction. We therefore report only the degraded-frame fraction, defined explicitly as $\Pr(D_t > 0)$.

References

[1] J. Jing, Y. Mao, A. Qiu, and K. Mikolajczyk, "Match Stereo Videos via Bidirectional Alignment," in *Proc. European Conf. Computer Vision (ECCV)*, 2024.

[2] H. Xu, C. Xu, and S. Giannarou, "StereoDiffusion: Temporally consistent stereo depth estimation with diffusion models," in *Proc. MICCAI*, 2024.

[3] F. Aleotti *et al.*, "Neural disparity refinement for arbitrary resolution stereo," in *Proc. Int. Conf. 3D Vision (3DV)*, 2021.

[4] W.-S. Lai *et al.*, "Learning blind video temporal consistency," in *Proc. European Conf. Computer Vision (ECCV)*, 2018.

[5] Z. Zeng *et al.*, "Temporally consistent stereo matching," in *Proc. European Conf. Computer Vision (ECCV)*, 2024.